Thomas Nguyen

Problem 1

Part (a): Building the Tree

Root entropy (5 enjoy, 6 don't, out of 11):d

$$H(S) = -\frac{5}{11}\log_2\frac{5}{11} - \frac{6}{11}\log_2\frac{6}{11} = 0.9927$$

Information Gain for each attribute:

Type (Fast Food: 1+/2−, Ethnic: 2+/1−, Casual: 2+/3−):

$$IG = 0.9927 - \left(\frac{3}{11}(0.9183) + \frac{3}{11}(0.9183) + \frac{5}{11}(0.9710)\right) = 0.0505$$

$$IG = 0.9927 - \left(\frac{4}{11}(1.0) + \frac{5}{11}(0.9710) + \frac{2}{11}(0)\right) = 0.1877$$

Neighborhood (Oakland: 1+/2−, SqHill: 2+/2−, Shadyside: 2+/2−):

$$IG = 0.9927 - \left(\frac{3}{11}(0.9183) + \frac{4}{11}(1.0) + \frac{4}{11}(1.0)\right) = 0.0151$$

Restriction (None: 2+/2−, Veg: 2+/2−, GF: 1+/2−):

$$IG = 0.9927 - \left(\frac{4}{11}(1.0) + \frac{4}{11}(1.0) + \frac{3}{11}(0.9183)\right) = 0.0151$$

Best root split: Price (IG = 0.1877)

{R4(0), R9(0)} → Leaf: **NO**

Price = $R1(0), R5(1), R7(1), R8(0)$
H = 1.0
Best split: Type (IG = 0.5)

- Type = Fast Food → {R1(0)} → Leaf: **NO**
- Type = Ethnic → {R7(1)} → Leaf: **YES**
- Type = Casual → {R5(1), R8(0)} → Split on Neighborhood
  - Neighborhood = Oakland → {R5(1)} → Leaf: **YES**
  - Neighborhood = Shadyside → {R8(0)} → Leaf: **NO**

Price = $R2(0), R3(0), R6(1), R10(1), R11(1)$
H = 0.9710
Best split: Neighborhood (IG = 0.4200)

- Neighborhood = Shadyside → {R10(1), R11(1)} → Leaf: **YES**
- Neighborhood = Squirrel Hill → {R2(0), R3(0), R6(1)} → Split on Type
  - Type = Ethnic → {R2(0)} → Leaf: **NO**
  - Type = Casual → {R3(0)} → Leaf: **NO**
  - Type = Fast Food → {R6(1)} → Leaf: **YES**

Classify R12–R16

Restaurant	Type	Price	Neighborhood	Path	Prediction
R12	Fast Food	$	Squirrel Hill	$ → Fast Food	NO
R13	Ethnic	$$	Shadyside	$$ → Shadyside	YES
R14	Fast Food	$	Oakland	$ → Fast Food	NO
R15	Casual	$$	Shadyside	$$ → Shadyside	YES
R16	Ethnic	$	Squirrel Hill	$ → Ethnic	YES

Recommendation: Go to R13 or R15 (both predicted YES)

Part (c):

Restaurant	Actual	Predicted	Result
R12	0	0	TN
R13	1	1	TP
R14	0	0	TN
R15	1	1	TP
R16	0	1	FP

TP = 2, TN = 2, FP = 1, FN = 0

$$E_{test} = \frac{1}{5} = 0.2$$

$$\text{Precision} = \frac{2}{2+1} = 0.6667$$

$$\text{Recall} = \frac{2}{2+0} = 1.0$$

$$F1 = 2 \cdot \frac{0.6667 \times 1.0}{0.6667 + 1.0} = 0.8$$

The only error (R16) occurs because the tree learned from R7 (Squirrel Hill example. It can't distinguish by restriction type at that leaf. This is a case of overfitting a small training set.

Problem 2

From 25 days: **YES = 12, NO = 13**

Conditional counts:

	Sunny	Overcast	Rain	Weak	Strong
YES	4	4	4	8	4
NO	5	5	3	8	5

$$P(\text{Yes}) = \frac{12}{25}, \quad P(\text{No}) = \frac{13}{25}$$

Day 26: Sunny, Strong

$$P(\text{Yes}) \propto \frac{12}{25} \cdot \frac{4}{12} \cdot \frac{4}{12} = 0.0533$$

$$P(\text{No}) \propto \frac{13}{25} \cdot \frac{5}{13} \cdot \frac{5}{13} = 0.0769$$

Answer: NO (0.0769 > 0.0533)

Day 27: Overcast, Weak

$$P(\text{Yes}) \propto \frac{12}{25} \cdot \frac{4}{12} \cdot \frac{8}{12} = 0.1067$$

$$P(\text{No}) \propto \frac{13}{25} \cdot \frac{5}{13} \cdot \frac{8}{13} = 0.1231$$

Answer: NO (0.1231 > 0.1067)

Day 28: Rain, Weak

$$P(\text{Yes}) \propto \frac{12}{25} \cdot \frac{4}{12} \cdot \frac{8}{12} = 0.1067$$

$$P(\text{No}) \propto \frac{13}{25} \cdot \frac{3}{13} \cdot \frac{8}{13} = 0.0739$$

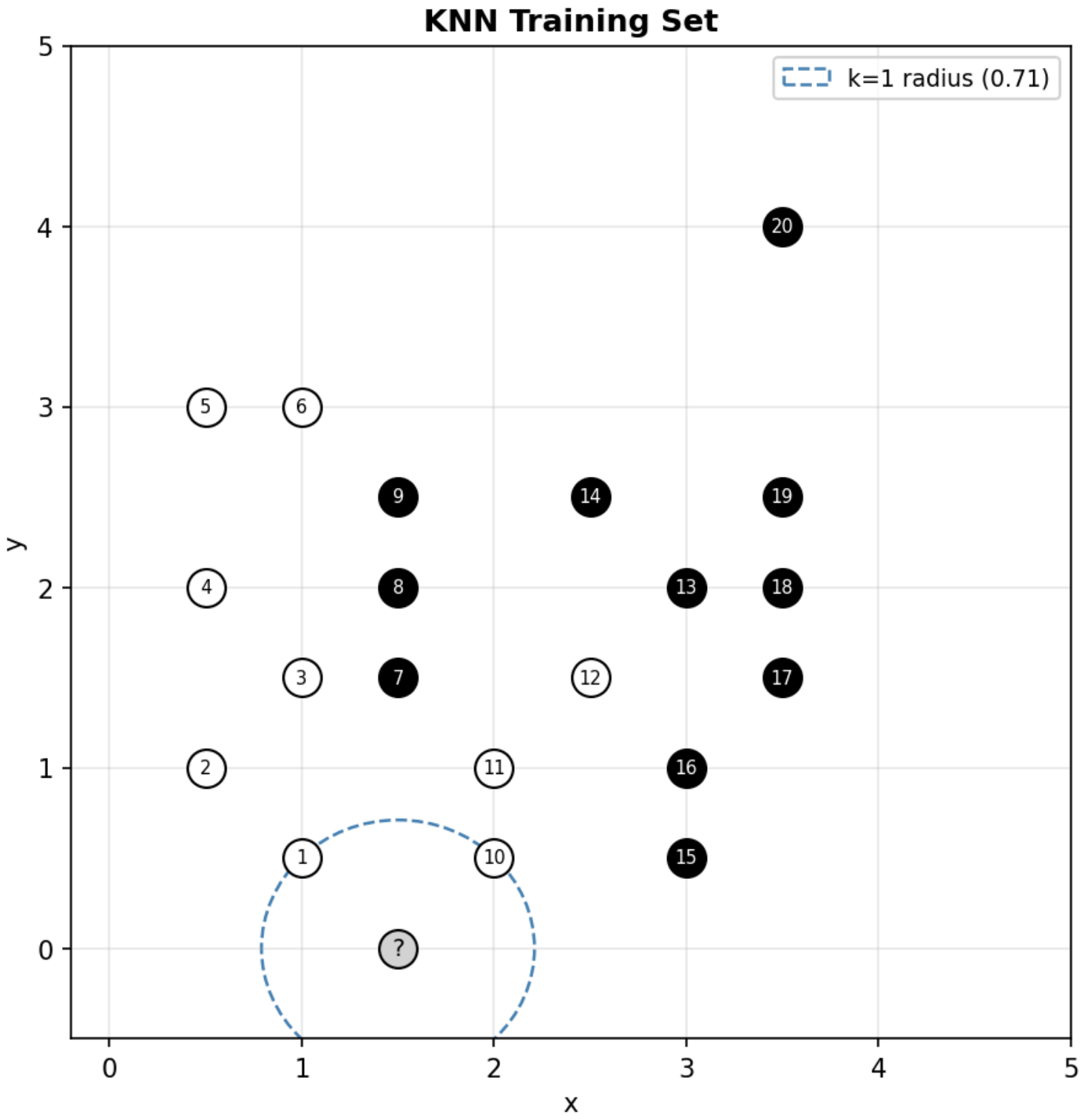
Answer: YES (0.1067 > 0.0739)

Problem 3: KNN

Point coordinates from Figure 1:

Point	x	y	Class
1	1	0.5	White

Point	x	y	Class
2	0.5	1	White
3	1	1.5	White
4	0.5	2	White
5	0.5	3	White
6	1	3	White
7	1.5	1.5	Black
8	1.5	2	Black
9	1.5	2.5	Black
10	2	0.5	White
11	2	1	White
12	2.5	1.5	White
13	3	2	Black
14	2.5	2.5	Black
15	3	0.5	Black
16	3	1	Black
17	3.5	1.5	Black
18	3.5	2	Black
19	3.5	2.5	Black
20	3.5	4	Black
?	1.5	0	?



Part 1: Classify "?" at (1.5, 0)

Nearest neighbors by Euclidean distance:

Point	Distance	Class
1 (1, 0.5)	0.7071	White
10 (2, 0.5)	0.7071	White
11 (2, 1)	1.118	White

- k=1: nearest is Point 1 (or 10) → **White**
- k=2: Points 1 and 10, both White → **White**
- k=3: Points 1, 10, 11, all White → **White**

Part 2: Misclassified points with k=1

For k=1, a training point is misclassified if its nearest neighbor has the opposite class.

Point 7 (1.5, 1.5) is Black. Its nearest neighbor is Point 3 (1, 1.5) at distance 0.5, which is White. So Point 7 would be predicted White — **misclassified**.

All other points have nearest neighbors of the same class. **Only Point 7 is misclassified with k=1.**

Part 3: Custom distance metric for k=1

Looking at the data, Black points are concentrated at higher x-values while White points sit at lower x-values. The issue is that Point 7 (1.5, 1.5) is horizontally close to White points but is actually Black.

Proposed metric:

$$d(a,b) = \sqrt{4(x_a - x_b)^2 + (y_a - y_b)^2}$$

By scaling the x-dimension by a factor of 2, horizontal separation dominates. For Point 7, its nearest neighbor becomes Point 8 (1.5, 2.0) — same x, distance = 0.5 — rather than Point 3 (1, 1.5) — Δx = 0.5 scaled → effective distance = 1.0. This correctly classifies all training points.

Part 4: k=5 with the weighted metric

With the weighted metric emphasizing horizontal separation, k=5 still classifies most points correctly since each point's 5 nearest neighbors stay predominantly within their horizontal region. However, boundary points like Point 7, 8, and 9 (at x = 1.5) may experience more instability under k=5 as majority voting across 5 neighbors can pull toward the larger White cluster on the left. The result is similar to k=1 for most points, but accuracy at the boundary may decrease slightly.

Problem 4: Hierarchical Clustering

Distance matrix (d = 1 − similarity):

d	P1	P2	P3	P4	P5	P6
P1	0	0.30	0.35	0.60	0.80	0.95
P2	0.30	0	0.05	0.30	0.50	0.65
P3	0.35	0.05	0	0.25	0.45	0.60
P4	0.60	0.30	0.25	0	0.20	0.35

d	P1	P2	P3	P4	P5	P6
P5	0.80	0.50	0.45	0.20	0	0.15
P6	0.95	0.65	0.60	0.35	0.15	0

MIN (Single Linkage)

P2–P3 = 0.05 → merge {P2, P3}. Updated (min): {P2,P3}–P1=0.30, P4=0.25, P5=0.45, P6=0.60

P5–P6 = 0.15 → merge {P5, P6}. Updated (min): {P5,P6}–P1=0.80, {P2,P3}=0.45, P4=0.20

P4–{P5,P6} = 0.20 → merge {P4, P5, P6}. Updated (min): {P4,P5,P6}–P1=0.60, {P2,P3}=0.25

{P2,P3}–{P4,P5,P6} = 0.25 → merge {P2,P3,P4,P5,P6}. Updated: to P1=0.30

P1–{P2,P3,P4,P5,P6} = 0.30 → merge all

MAX (Complete Linkage)

P2–P3 = 0.05 → merge {P2, P3}. Updated (max): {P2,P3}–P1=0.35, P4=0.30, P5=0.50, P6=0.65

P5–P6 = 0.15 → merge {P5, P6}. Updated (max): {P5,P6}–P1=0.95, {P2,P3}=0.65, P4=0.35

{P2,P3}–P4 = 0.30 → merge {P2, P3, P4}. Updated (max): {P2,P3,P4}–P1=0.60, {P5,P6}=0.65

P1–{P2,P3,P4} = 0.60 → merge {P1,P2,P3,P4}. Updated: {P1,P2,P3,P4}–{P5,P6}=0.95

{P1,P2,P3,P4}–{P5,P6} = 0.95 → merge all

